

PHAN VAN BANG

AI Engineer

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SUMMARY

AI Engineer with 2+ years of professional experience specializing in Multi-Agent Systems, RAG pipelines, Voice AI, and IoT integration. Built 56+ production AI projects spanning 8+ business domains, including a multi-agent orchestration system with 94+ MCP tools, Swarm Intelligence, and continual learning capabilities.

TECHNICAL SKILLS

Languages:	Primary: Python, TypeScript/JavaScript Proficient: C#, Java, Go Familiar: Rust, C++
AI & LLM:	LLM, Fine-tuning (LoRA/QLoRA), RAG & GraphRAG, Multi-Agent Orchestration, Swarm Intelligence, MCP Protocol (94+ tools), Continual Learning, vLLM, ComfyUI, HuggingFace
CV & NLP:	PaddleOCR, Stable Diffusion, ComfyUI, Face Recognition, spaCy (vi/en), BGE Embeddings
Voice AI:	OpenAI Whisper STT, TTS (OpenAI/Fish Audio/ElevenLabs), VAD, Full-Duplex Voice Chat
Frameworks:	LangChain, LlamaIndex, FastAPI, Streamlit, React, Next.js, CrewAI, AutoGen, Unsloth
Databases:	PostgreSQL + pgvector, Qdrant, MongoDB, Redis, Neo4j, UMAP
Architecture:	Microservices, RESTful API, WebSocket, SSE Streaming, Docker Compose, Celery + Redis, MCP Client+Server (OAuth 2.1 PKCE), MQTT
DevOps & Cloud:	AWS (S3, EC2), Docker Hub, GitHub Actions, Nginx, RunPod (GPU Serverless), Let's Encrypt SSL
Tools:	Git, N8N Automation, Data Crawling/Scraping, Playwright

WORK EXPERIENCE

AI Engineer MekongAI

06/2023 - Present

- > Designed & built multi-agent orchestration system (MekongAI Nexus) with 94+ MCP tools, Swarm Intelligence (debate, consensus, hierarchical, parallel), and XSkill continual learning, serving 500+ daily conversations
- > Developed AI learning content platform (Knowlympic) with OCR pipeline, automated question generation, reducing content creation time by 80% for 3 English language centers
- > Built AI interaction platform (MetaHuman) with personality engine, 3-tier memory system, full-duplex voice, and 3D avatar integration, achieving <500ms voice response latency
- > Integrated AI with ESP32 hardware for voice-interactive IoT bots via MCP protocol
- > Implemented GraphRAG memory with pgvector + hybrid retrieval (vector + FTS + graph RRF), improving retrieval accuracy by 35% over baseline vector search
- > Designed 13 security layers (CommandGuard, SSRF Guard, PII Redaction, etc.)
- > Built multi-channel deployment: WebSocket, Telegram, Discord, Slack, WhatsApp, Email, handling 1000+ concurrent connections
- > Built and maintained 56+ AI projects spanning 8+ business domains (Legal, Education, Recruitment, POS, Crypto, Social Media, Content Generation, IoT)
- > Fine-tuned Llama 3.1 models (8B/70B) using Unsloth framework with trajectory compression → SFT data pipeline
- > Built 20+ ComfyUI AI image processing workflows, processing 5000+ images/month for clients
- > Created real-time crypto monitoring system with MA crossover signals and automated Telegram alerts (Binance API)
- > Built face similarity API and computer vision pipelines (SAM3, PaddleOCR-VL) deployed on RunPod GPU serverless

KEY PROJECTS

1. MekongAI Nexus — Multi-Agent Orchestration System

- > Self-spawning agent system with Swarm Intelligence (4 strategies: debate, consensus, hierarchical, parallel)
- > 94 MCP tools across 28+ categories, GraphRAG memory, XSkill continual learning
- > 13 security layers, 9 communication channels, Docker Hub deployed

Tech: FastAPI, React, PostgreSQL + pgvector, Redis, Multi-Provider LLM, Docker

2. MekongAI Knowlympic — AI Learning Content Platform

- > AI-powered platform auto-generating learning content from textbook PDFs via hybrid OCR
- > Auto test generator (multi-version A/B/C/D), RAG-based AI chat assistant

Tech: FastAPI, React, Celery + Redis, Qdrant, OpenAI, PaddleOCR

3. MekongAI MetaHuman — AI Personality Platform

- > AI with personality engine (5D Drive System, Emotion Critic, Inner Monologue)
- > 3-Tier Memory (Hot/Warm/Cold), full-duplex voice, 3D avatar with 10 emotion states

Tech: FastAPI, React, PostgreSQL, OpenAI Whisper/TTS, Fish Audio

4. ESP32 AI Bot — IoT Voice Assistant

- > Voice-interactive AI bot on ESP32 hardware with MCP protocol integration
- > Voiceprint recognition, visual perception, IoT device control via MQTT

Tech: Python, ESP32-S3, MQTT, MCP, FunASR, Multi-LLM

5. AI Recruitment Platform — Vietnamese NLP Matching

- > AI-powered CV parsing + Vietnamese NLP job-candidate matching
- > Vector similarity search with spaCy dual pipeline (vi + en)

Tech: FastAPI, Qdrant, spaCy, OCR, AWS S3

6. AI Content Generator — Multi-Platform

- > Multi-agent pipeline for auto-generating content (Facebook, Instagram, TikTok)
- > RAG-based brand voice learning from existing content

Tech: CrewAI, Streamlit, Qdrant, RAG, Multi-LLM

EDUCATION

B.Eng. in Information Technology

Da Nang, 2020 - 2025

Vietnam-Korea University of Information & Communication Technology (VKU)

CERTIFICATIONS & LANGUAGES

- > English: Professional working proficiency (Reading technical docs, Writing, Speaking)
- > Deep Learning Specialization, LangChain & LLM Application Development
- > LLM Fine-tuning (Unsloth, LoRA/QLoRA) — Production deployed

SOFT SKILLS

- > Self-learning & Rapid adaptation
- > Team collaboration & technical leadership
- > Problem decomposition & analytical thinking
- > Ownership mindset & end-to-end delivery